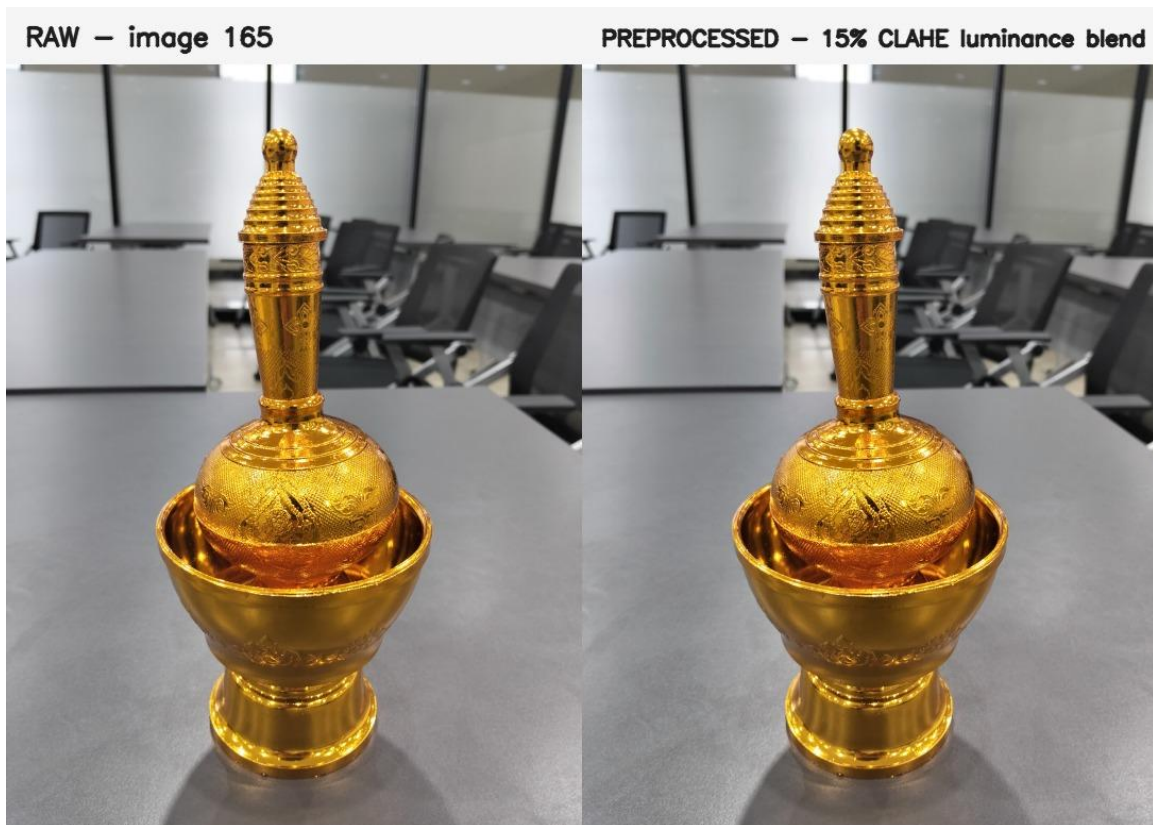


CSX4213 Computer Vision Project

Thai Libation Brass Vessel 3D Reconstruction

Verified Image QA and Preprocessing Progress Walkthrough



Representative capture and the selected mild preprocessing variant.

Prepared by: Sithu Win San, Eaint Myat Thu, and Gulizara Benjapalaporn

Assumption University | CSX4213 Computer Vision

Verified 27 August 2026

Executive Summary

PREPROCESSING COMPLETE: 288 evidence-selected, geometry-preserving images are ready for pyCOLMAP; all 297 originals still match their baseline hashes.

The project audited 297 OPPO Reno12 F photographs of a Thai brass libation vessel, calibrated quality warnings from the real capture distribution, retained useful coverage, compared RAW and PREPROCESSED neighboring-frame correspondences, and exported an unambiguous next-stage image set. The phase stops before pyCOLMAP reconstruction.

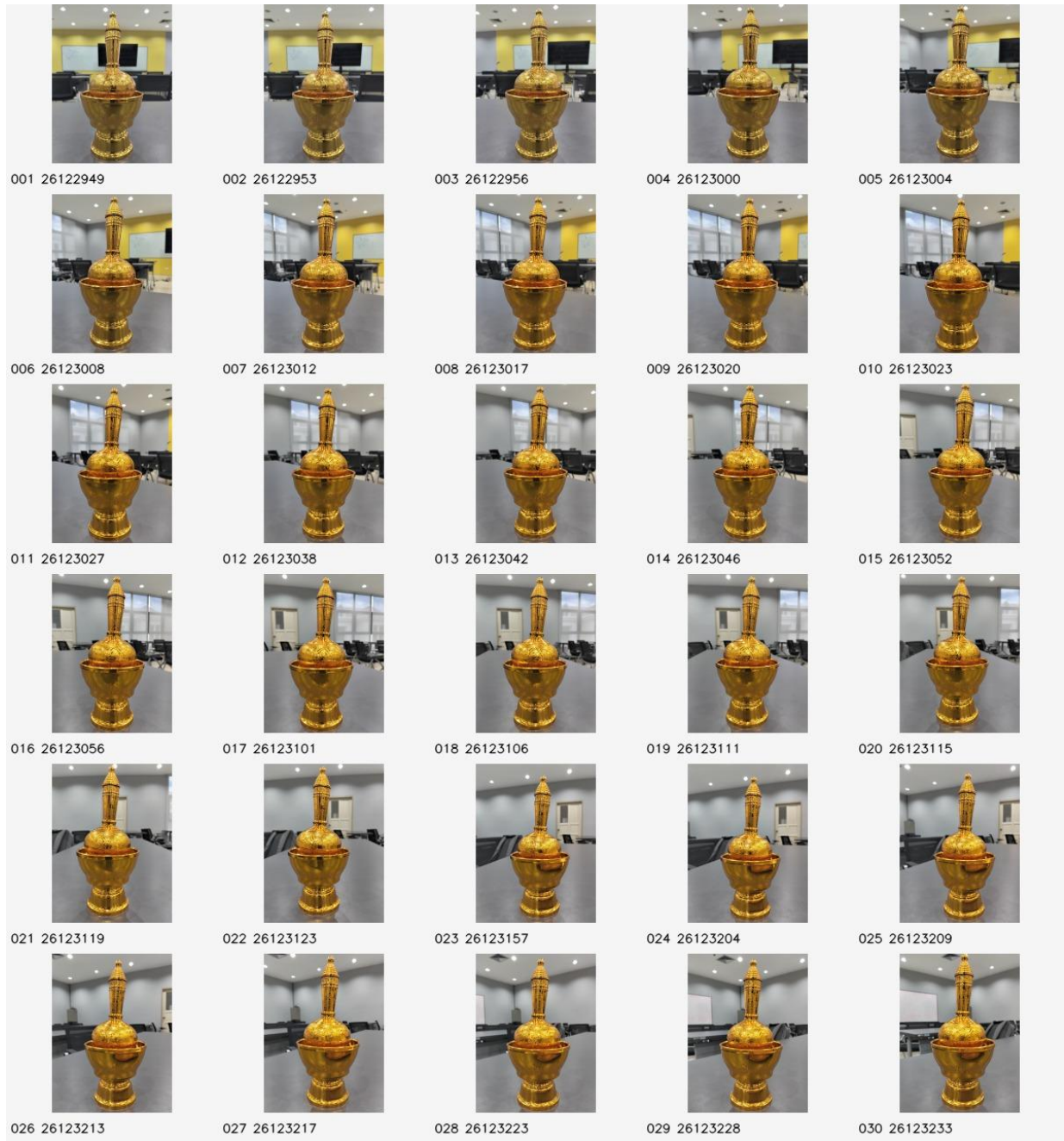
Evidence item	Verified result
Raw captures	297 readable JPEGs, 3072 x 4080
Final decisions	207 ACCEPT 81 WARN 9 REJECT
Selected dataset	288 PREPROCESSED images
SIFT/RANSAC evidence	2,483 PREPROCESSED vs 2,376 RAW inliers
Pair consistency	PREPROCESSED non-worse on 9 of 10 pairs
Final integrity	0 raw mismatches 0 output duplicates
Hard boundary	pyCOLMAP has not started

1. Project Objective

Reconstruct a real Thai brass libation vessel from smartphone images through an explainable computer vision workflow: image QA and preprocessing, pyCOLMAP Structure from Motion, later dense reconstruction, meshing/texturing, and Blender cleanup. This report covers only the completed preprocessing boundary.

2. Capture Methodology

The 297 portrait photographs were captured at 3072 x 4080 with an OPPO Reno12 F. The main fixed-object sequence provides dense middle, low, elevated, top-down, and detail coverage with neighboring overlap. Images 289-297 form a later hand-held/flipped sequence where object pose and background relation change and hands occlude the vessel.



Example contact sheet from the full ten-sheet sequence review.

3. Why the Originals Are Preserved

The raw directory IMG20260826122949/ is immutable source evidence. Processing reads from it and writes only to preprocessing/. raw_manifest_before.json records the expected filename, byte count, and SHA-256 digest of every original, making accidental mutation observable.

4. The 297-Image Audit

- 297 of 297 JPEGs readable; consistent 3072 x 4080 dimensions, OPPO Reno12 F metadata, and EXIF orientation 1.
- No exact SHA-256 duplicates and no conservative adjacent dHash near-duplicate candidates.

- All ten original contact sheets visually reviewed across the entire acquisition sequence.
- Existing audit reports were reused; obsolete exploratory scripts were not recreated.

5. Image-Quality Metrics and Real-Data Thresholds

quality_check.py measures an 800-pixel-wide in-memory analysis copy: luminance mean, contrast, Laplacian-variance sharpness, dark/bright clipping, and SIFT feature count. Thresholds are the eligible dataset's 5th/95th percentiles plus conservative severe components.

Threshold	Final value	Purpose
Sharpness warning	178.1662	Review low-detail frames
Severe sharpness component	131.3283	Reject only with severe low features
Brightness low / high	120.7684 / 139.3575	Dataset-relative exposure review
Contrast warning	31.1619	Review low-separation views
Dark / bright clip (%)	0.1017 / 0.8486	Review tonal extremes
Feature count warning	1,320	Review weak local texture
Severe feature component	528	Reject only with severe blur

6. Visual and Contact-Sheet Inspection

After the final run, four decision sheets containing every WARN and REJECT case were opened. Warning frames remained useful views with expected brass highlights, angular continuity, and overlap. All ten before/after figures were also opened to check that the transform remained mild and geometry-preserving.

7. ACCEPT, WARN, and REJECT Reasoning

Decision	Count	Meaning	Included for pyCOLMAP?
ACCEPT	207	No dataset-relative review signal	Yes
WARN	81	Review signal; still useful coverage	Yes
REJECT	9	Fixed-object geometry break (289-297)	No



267 WARN

low brightness

268 WARN

low brightness

269 WARN

low brightness

270 WARN

low brightness

271 WARN

low brightness



285 WARN

low brightness

289 REJECT

separate hand-held/flipped

290 REJECT

separate hand-held/flipped

291 REJECT

separate hand-held/flipped

292 REJECT

separate hand-held/flipped



293 REJECT

separate hand-held/flipped

294 REJECT

separate hand-held/flipped

295 REJECT

separate hand-held/flipped

296 REJECT

separate hand-held/flipped

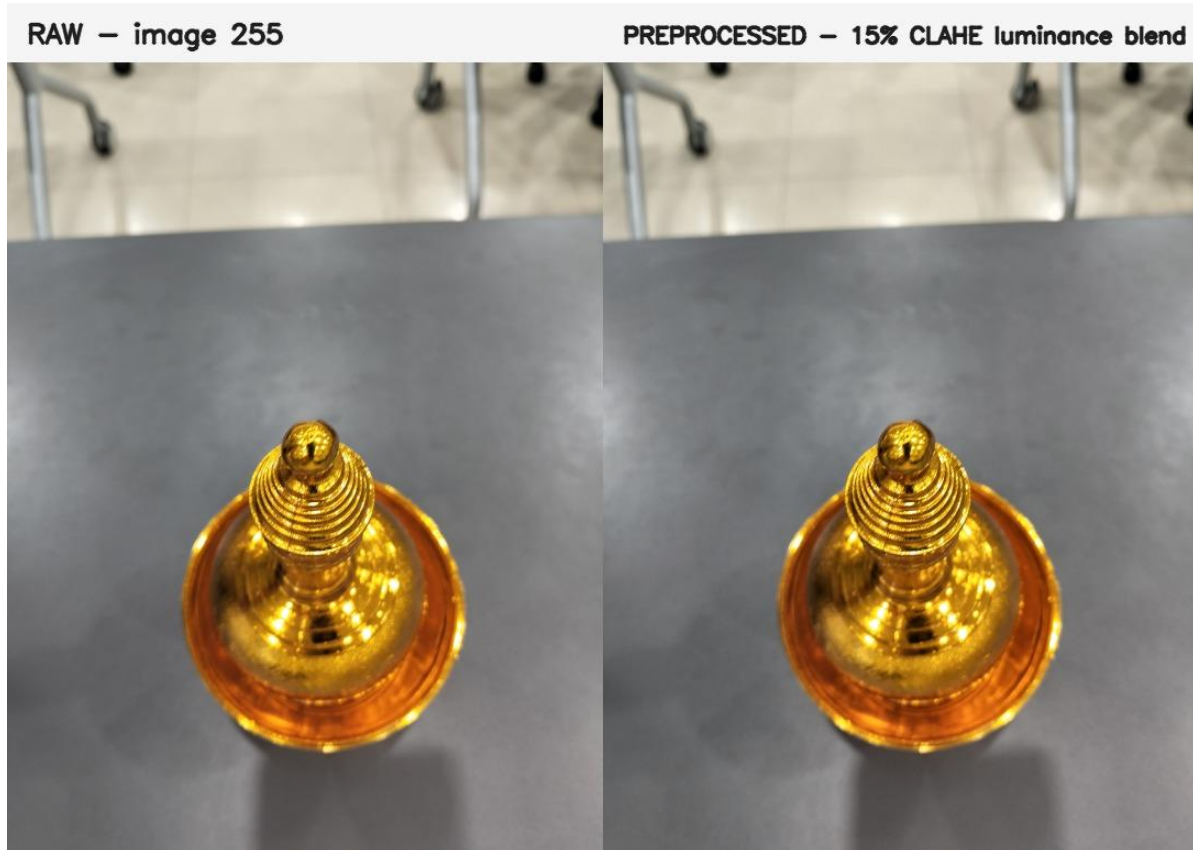
297 REJECT

separate hand-held/flipped

The nine red REJECT frames are the separate hand-held/flipped and hand-occluded sequence.

8. Conservative Preprocessing

preprocess_images.py converts BGR to LAB, applies CLAHE to luminance, blends 15% of the enhanced luminance into the original, and converts back to BGR. It never crops, rotates, resizes, warps, perspective-corrects, synthesizes detail, or removes reflections. Outputs are deterministic uint8 BGR images at 3072 x 4080.



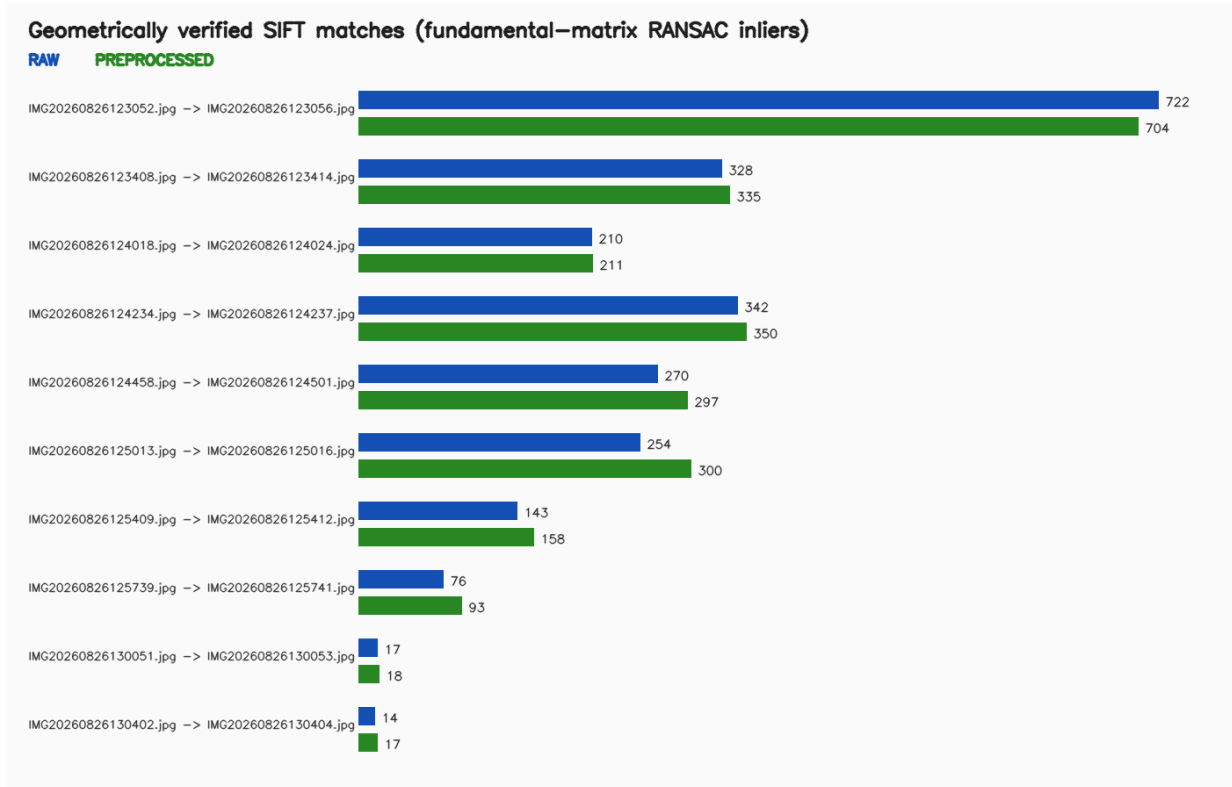
Low-feature top-down case: subtle photometric change, identical geometry.

9. RAW vs PREPROCESSED SIFT Experiment

Ten neighboring pairs sample early, middle, elevated, top-down, low-feature, and late sequence regions. Both variants use OpenCV SIFT (up to 8,000 features at maximum width 1,200), BFMatcher with L2 distance, two neighbors, Lowe ratio 0.75, and fundamental-matrix RANSAC (1.5 px, 0.99 confidence). PREPROCESSED matching decodes the exact deterministic quality-95 JPEG bytes used by final export.

Pair indices	RAW good / inliers	PREPROCESSED good / inliers
15-16	905 / 722	958 / 704
45-46	478 / 328	505 / 335
75-76	365 / 210	362 / 211
105-106	466 / 342	497 / 350
135-136	468 / 270	483 / 297
165-166	451 / 254	478 / 300

195-196	272 / 143	290 / 158
225-226	228 / 76	238 / 93
255-256	49 / 17	57 / 18
280-281	61 / 14	67 / 17
TOTAL INLIERS	2,376	2,483



PREPROCESSED produced at least as many verified inliers on nine of ten tested pairs.

10. Evidence Used to Choose the Input

The selection rule was fixed before reading the outcome: choose PREPROCESSED only when total verified inliers are greater and it is non-worse on at least half the pairs; otherwise preserve RAW. PREPROCESSED produced 107 additional inliers (4.50% more) and was non-worse on 9 of 10 pairs, so it became the reconstruction input.

11. Final pyCOLMAP-Ready Dataset

- Exact directory: preprocessing/pycolmap_input/images/
- 288 JPEGs using deterministic original capture filenames.
- Every output reopened successfully at 3072 x 4080.
- Every output hash matches selection_manifest.csv; zero duplicates found.
- All ACCEPT and WARN images included; all nine REJECT images absent.

12. Hash-Based Proof of Raw Immutability

The workflow verifies the baseline before processing and again afterward. A separate final integrity audit then re-hashed every raw file and reopened every selected output.

Verification	Scope	Outcome
Python compile	3 preprocessing scripts	PASS
Focused tests	15 behavior/integration tests	15 PASS, 0 FAIL
Real pipeline	297 raw images	Exit 0
Selected outputs	288 images	288 readable, all 3072 x 4080
Selection hashes	288 manifest entries	0 mismatches, 0 duplicates
Raw re-hash	297 baseline entries	0 size/SHA-256 mismatches
Visual review	10 before/after + 4 decision sheets	Opened and inspected

13. What Comes Next in pyCOLMAP

In a separate phase, verify the installed pyCOLMAP API and environment, point it only at the final 288-image directory, extract features, match views, run incremental mapping, and evaluate registered-image count, reprojection error, track length, and coverage before dense reconstruction.

STOPPING BOUNDARY: no pyCOLMAP, sparse/dense reconstruction, meshing, texturing, or Blender reconstruction work has run yet.

Professor Demonstration: Step-by-Step Speaking Guide

1. Open README.md and explain the full smartphone-capture to 3D-model pipeline.
2. Show IMG20260826122949/ and raw_manifest_before.json as immutable source evidence.
3. Open early and late contact sheets to demonstrate angular coverage and overlap.
4. Open quality_decisions.csv and explain the measured columns and three decisions.
5. Show the real-data threshold table and explain why warnings are not automatic rejects.
6. Open the four decisions_*.jpg sheets and point out that useful warning frames remain included.
7. Use decisions_076_090.jpg to explain the fixed-object geometry break in images 289-297.
8. Open two before_after_*.jpg figures and explain the 15% luminance-only CLAHE blend.
9. Open sift_inliers.png and explain SIFT, the 0.75 ratio test, and RANSAC inliers.
10. Explain the predeclared selection rule and the 2,483 versus 2,376 result.
11. Show preprocessing/pycolmap_input/README.md and the 288-image images/ directory.
12. Open raw_verification_after.json and state that all 297 baseline hashes still match.
13. Finish clearly: preprocessing is verified; pyCOLMAP reconstruction is the next stage, not a completed claim.

Limitations and Honest Claims

- The ten-pair experiment is representative evidence, not an all-pairs benchmark.
- SIFT/RANSAC improvement does not prove that every selected image will register in Structure from Motion.
- All 81 WARN frames are deliberately retained; pyCOLMAP diagnostics may justify later targeted removal.
- No sparse model, dense model, mesh, texture, or final 3D reconstruction exists yet.